

Max Marks = 10

Note : Answers in the space provided or tick the box with correct option. For rough work use the blank side.

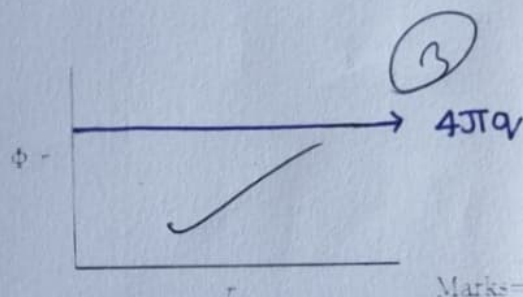
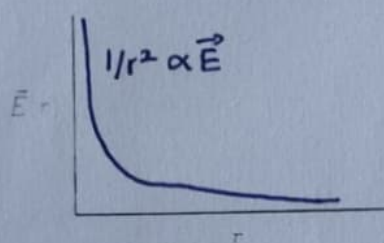
1. (a) ☒ King's experiment is aimed at finding a bound on the difference between proton charge and electron charge.
 (b) ☐ King's experiment is aimed at verifying Coulomb's law.
 (c) ☐ King's experiment is aimed at measuring the relativistic invariance of charge.
 (d) ☐ King's experiment measures electrostatic energy stored in the atoms.

Marks = 2

2. (a) ☐ Cavendish experiment is aimed at finding a bound on difference between the proton charge and electron charge.
 (b) ☒ Cavendish experiment is aimed at verifying Coulomb's law.
 (c) ☐ Cavendish experiment is aimed at measuring the relativistic invariance of charge.
 (d) ☐ Cavendish experiment measures electrostatic energy stored in the atoms.

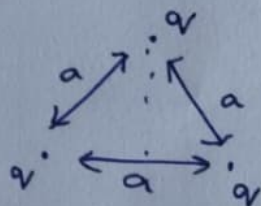
Marks = 2

3. Consider a point charge q at the origin and r is the distance from the origin. $\Phi(r)$ is the flux passing through a sphere of radius r centered at the origin. Complete the plots.



Marks = 3

4. Three charges of magnitude q are placed on an equilateral triangle. If the side of the triangle is a , what is the energy of this configuration? How does the energy change as a function of the area of the triangle?



Work done in bringing the charges individually from infinity to the configuration } = Energy of configuration (say 'C')

$$C = 0 + \left[\frac{q^2}{a} - 0 \right] + \left[\frac{q^2}{a} + \frac{q^2}{a} - 0 \right]$$

(A) area = $\frac{1}{2} \times a \times (h)$

$A = \frac{1}{2} \times a \times \frac{\sqrt{3}a}{2}$
 $A = \frac{\sqrt{3}a^2}{4}$
 $\sin 60^\circ = \frac{\sqrt{3}}{2} = \frac{h}{a}$

$$C = \frac{3q^2}{a}$$

using work energy theorem.

Marks = 3

$$\sqrt{\frac{4A}{\sqrt{3}}} = a$$

$$C = \frac{3q^2 \sqrt{3}}{2\sqrt{A}}$$

3